EE4-17
EE9-AO4

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2021

MSc and EEE PART IV: MEng and ACGI

HIGH PERFORMANCE ANALOGUE ELECTRONICS

Tuesday, 18 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : A.S. Holmes

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The Questions

1.

(a) Briefly explain in your own words the advantages of a double superheterodyne receiver with respect to a simple superheterodyne (Note: Do not copy or rephrase from the lecture notes. That will not be considered "your own words").

[4]

(b) Briefly explain with your own words the reasons why a designer might put a filter after a local oscillator in a superheterodyne receiver (Note: Do not copy or rephrase from the lecture notes. That will not be considered "your own words").

[4]

(c) In a receiver, a bandpass filter is used to remove image signals located in a 10MHz to 12MHz band. What is the frequency of the local oscillator if the RF signal of interest is located between 7MHz and 10MHz?

[4]

(d) In a receiver, the local oscillator is oscillating at 32MHz. Would a circuit with the transfer function shown below contribute to attenuate the problem of spurious harmonics? Explain very briefly your answer.

$$\frac{5.685\cdot 10^8}{s^2+1.066\cdot 10^4 s+5.685\cdot 10^8}$$

[4]

(e) The noise factor of a cascade of two circuit blocks is given by:

$$F = F_1 + (F_2-1)/G_1$$

where the subscript numbers refer to the order of the block in the cascade and G is the power gain. What is the expression for the noise factor of a cascade of four circuit blocks?

[4]

2.

(a) Draw the schematic of a differential pair with a current mirror as a load.

[2]

(b) Draw the different noise sources in the circuit, at the transistor level (i.e. there is no need to draw a small signal schematic).

[2]

(c) Explain which noise sources are most important and which ones can be neglected.

[2]

(d) Is a differential configuration better or worse in terms of noise than a single order one? Briefly explain your answer.

[2]

(e) Is a differential configuration better or worse in terms of distortion than a single order one? Briefly explain your answer.

[2]

(f) Explain a reason why a fourth order harmonic might be present at the output of a differential circuit after fabrication.

[2]

(g) Derive the expression for the current at the output of a differential pair with degeneration.

[2]

(h) For the circuit in (g) what design choice would you make to increase linearity? Would this design choice have any negative effect? Briefly explain your answer.

[3]

(i) Assuming you design the circuit in (g) having very high linearity as a target, what would be the equivalent input noise of the circuit? (Note: Neglect all the sources that would be small relative to the dominant ones).

[3]

3.

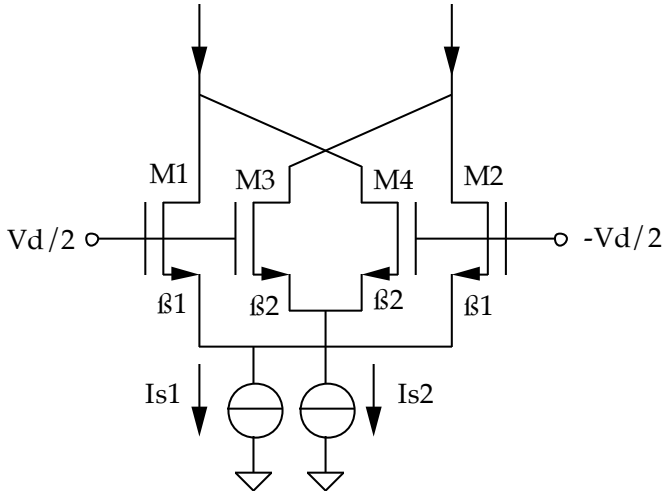
(a) Is the following statement correct? Briefly explain your answer.

"In a simple differential pair transconductor topology the transconductance is always approximately given as:

$$gm = \frac{\partial \text{Iout}}{\partial \text{Vin}} = \frac{\text{I}}{\text{Vt}} \text{sec h}^2\left(\frac{\text{Vin}}{2\text{Vt}}\right) \text{"}$$

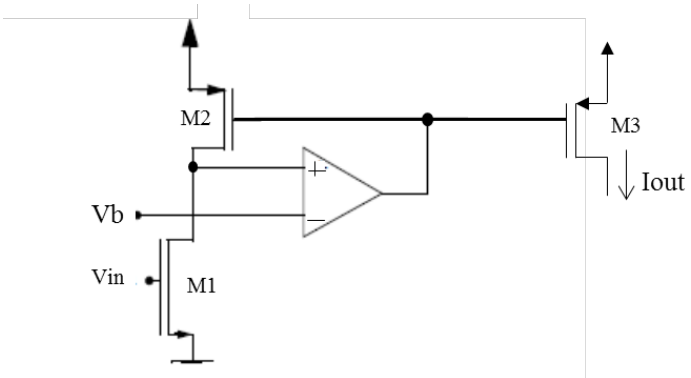
[4]

(b) The circuit below is sent for fabrication and when it is tested a significant distortion, coming from a third order harmonic, is observed. Assuming all the transistors are properly matched and operating in the most adequate region, what could the designer have done wrong that led to that distortion? (Note: Use either words or equations to explain your answer).



[4]

(c) The output current of the circuit below shows a strong second order harmonic. Is this expected? What do you think could be causing it?



[4]

(d) Explain two alternative ways of modifying the circuit in (c) to improve its linearity performance. (Note: you can draw a new circuit, but it has to be based in this fundamental block). [4]

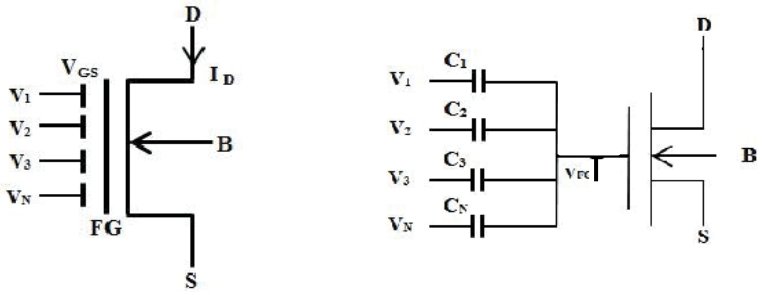
(e) A differential version of the circuit in (c) comes back from fabrication and it is found to be oscillating. Why do you think that might be? [4]

4.

(a) A FGMOS transistor is a multi-input MOS transistor with an array of capacitors connected to the gate, as shown in the figure below. Hence the voltage at the gate of the transistor is approximately given by:

$$V_G = V_{FG} \approx \sum_{i=1}^{N} \frac{C_i}{C_{Total}} V_i$$

where C_{Total} is the overall capacitance seen by the gate.



Assuming bulk and source are connected to ground, and the transistor is operating in strong inversion saturation, write an expression for the drain current in the transistor as a function of the voltages at the inputs (i.e. not as a function of the voltage at the gate).

[4]

(b) What are the two main types of noise in the FGMOS transistor? Write expressions for them referred to the gate.

[4]

(c) Assuming all the inputs, except for V1 are connected to constant biasing voltages, and V1 is the equivalent input, is the equivalent input noise in the FGMOS transistor better than, worse than or the same as in a normal MOS transistor? Briefly explain your answer.

[2]

(d) Assuming all the inputs, except for V1 are connected to constant biasing voltages, and V1 is the equivalent input, is the equivalent output noise in the FGMOS transistor better than, worse than or the same as in a normal MOS transistor? Briefly explain your answer.

[2]

(e) Assuming all the inputs, except for V1 are connected to constant biasing voltages, and V1 is the equivalent input, is the input range for the same level of distortion larger or smaller than in a normal MOS transistor?

[2]

(f) Invent a circuit in which you decide to use a FGMOS transistor instead of a MOS one. Explain what the purpose of that circuit is, and your design choices for that circuit. (For example, what tradeoffs are you making in terms of performance?)

[6]